

Name of Student ----- Enrolment No. -----

Department /School -----

BENNETT UNIVERSITY, GREATER NOIDA

End-Term Makeup Examination, Even Semester 2022-23

COURSE CODE: CSET206

MAX. DURATION:

2 Hour

COURSE NAME: Design and Analysis of Algorithms

COURSE CREDIT (L-T-P): 3 1 4

MAX. MARKS: 50

GENERAL INSTRUCTIONS: -

1. Do not write anything on the question paper except name, enrolment number and department/school.
2. Carrying mobile phone, smart watch and any other non-permissible materials in the examination hall is an act of UFM.

1. Find Time Complexity of the following fragment of the algorithms:

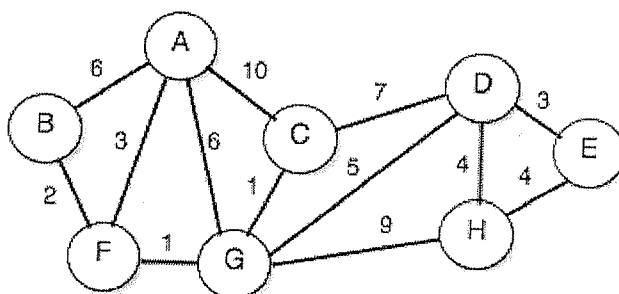
(1 + 1 + 1 + 2 Marks)

a) <pre>X=0; for (i=1; i<=n; i*=2) { for(j=1; j<=n; j++) X++; }</pre>	b) <pre>int a = 0, i = N; while (i > 0) { a += i; i /= 2; }</pre>
c) <pre>function fun(n) { for (int i = 2; i < n; ++i) { if (n % i == 0) { return false; } } return true; }</pre>	d) <pre>int recurSum(int n) { if (n <= 1) return n; return n + recurSum(n - 1); }</pre>

2. Write an algorithm to merge two sorted list and analyze its time complexity.

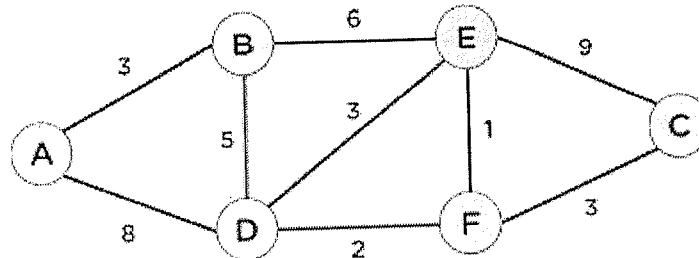
(2 + 1 Marks)

3. Differentiate between Kruskal and Prim algorithm for finding Minimal Spanning Tree of a given graph G. In the below given graph apply Prim algorithm to find Minimal Spanning Tree. (2 + 3 Marks)



4. Define the red-black tree with properties and construct a red-black tree with the given data. (Show each step of insertion). 2, 1, 4, 5, 9, 3, 6. (2 + 3 Marks)

5. Apply Dijkstra Algorithm for finding single source shortest path of the given graph, Starting from vertex A. Why Dijkstra Algorithm not preferred for negative edge weighted graph? (4 + 1 Marks)



6. Differentiate between the following: (3 X 2 Marks)
- Divide and Conquer and Dynamic Programming
 - B Tree and B + Tree
 - Hamiltonian cycle and Travelling Salesman problem
7. Give an algorithm for fractional knapsack problem using greedy approach. Whether your algorithm will provide optimal solution for 0/1 Knapsack problem (Explain your reason). (3 + 2 Marks)
8. Apply backtracking to solve 4 Queen problem. Draw the state space tree corresponding to 4 Queen problem. (4 + 4 Marks)
9. Differentiate between P, NP and NP Complete problems. List the 3 standard examples of NP Complete problems and also list 2 problems of NP hard class which are not in NP. (3 + 3 + 2 Marks)

😊😊😊 ALL THE BEST 😊😊😊